# Direct-Route Check of the Updated Shortcut Calculation

7 June 2026

## Purpose and conclusion

This note follows the original time-convolution route in the updated manuscript and performs the calculation explicitly. The result is a corrected direct-route version of the long Eq. (41) expansion, together with a check showing that the same expression recompresses to the closed form

$$\boxed{\widetilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}} \tag{1}$$

where

$$N = 2L, \qquad a = \frac{q}{\beta(1-q)}, \qquad y = 1 + \frac{1/z - 1}{q}, \qquad U_{-1} = 0.$$

Thus the closed form is not a different model or a shortcut around the calculation. It is the simplified form of the original convolution route after all intermediate poles have been summed.

## 1 Corrected starting point

Let

$$\lambda = \beta(1-q).$$

When the shortcut is $u \to v$ and $v$ is the absorbing target, the shortcut is killing in the transient state space. The corrected first-passage generating function is

$$\boxed{\widetilde{F}_{n_0}(v,z) = \widetilde{\mathcal{F}}_{n_0}(v,z) + qz\,\widetilde{W}_{n_0}(u,z)\left[1 - \widetilde{\mathcal{F}}_u(v,z)\right]\widetilde{H}(z)} \tag{2}$$

with

$$\widetilde{H}(z) = \frac{\beta(1-q)}{q}\,\frac{1}{1 + z\beta(1-q)\widetilde{W}_u(u,z)}. \tag{3}$$

For the antipodal case $|u - v| = N/2$, this is

$$\widetilde{H}(z) = \frac{1}{a + \phi(z)}. \tag{4}$$

The sign in (2) is positive. Therefore the corrected time-domain convolution is

$$\boxed{F_{n_0}(v,t) = \mathcal{F}_{n_0}(v,t) + q\sum_{t'=0}^{t-1} W_{n_0}(u, t-1-t') \sum_{t''=0}^{t'} \left[\delta_{t',t''} - \mathcal{F}_u(v, t'-t'')\right] H(t'')} \tag{5}$$

for $t \geq 1$. This is the corrected version of the updated manuscript's Eq. (40).

## 2 Mode expansions used in the direct route

For $N = 2L$, the no-shortcut first-passage probability is

$$\mathcal{F}_{n_0}(v,t) = \sum_{\ell=1}^{L} f_\ell(n_0,v)\alpha_\ell^{t-1}, \qquad t \geq 1. \tag{6}$$

The killed propagator is

$$W_{n_0}(u,t) = \sum_{r=1}^{N-1} G_r^{(1)}(n_0,u,v)\gamma_r^t + \sum_{r=1}^{L} g_r^{(2)}(n_0,u,v)\alpha_r^t. \tag{7}$$

Here

$$\alpha_r = 1 - q + q\cos\frac{(2r-1)\pi}{N}, \qquad \gamma_r = 1 - q + q\cos\frac{2\pi r}{N}.$$

The corrected $H$-poles are the roots of the corrected denominator. Write

$$s = 1/z, \qquad y_s = 1 + \frac{s-1}{q}.$$

For the antipodal case

$$\widetilde{H}(z) = \frac{T_L(y)}{aT_L(y) + U_{L-1}(y)}. \tag{8}$$

Define

$$D(y) = aT_L(y) + U_{L-1}(y).$$

Let

$$h_0 = H(0) = \frac{\beta(1-q)}{q} = \frac{1}{a}.$$

Since $\widetilde{H}(z)$ has numerator and denominator of the same degree, its partial-fraction expansion has a constant term. Equivalently, $H(0)$ must be kept separate. If $D(y_k) = 0$ and $s_k = 1 - q + qy_k$, then for $t \geq 1$

$$H(t) = \sum_k c_k s_k^{t-1}, \qquad \boxed{c_k = \frac{qT_L(y_k)}{D'(y_k)}}. \tag{9}$$

This residue formula is the safest way to write the corrected version of Eq. (39): it is tied directly to the corrected denominator $D(y)$. In practice one should solve $D(y) = 0$ itself, not only the trigonometric parametrization $y = \cos\theta$. Some roots can have $y < -1$ while still giving an admissible $|s_k| < 1$ pole after $s_k = 1 - q + qy_k$.

## 3 The convolution kernels

Equation (5) contains two kinds of sums, and each must be split into the $H(0) = h_0$ contribution and the positive-time $H(c)$, $c \geq 1$, pole contribution.

### Two-factor convolution

The $H(0)$ part gives simply

$$A_t^{(0)}(x) = x^{t-1}. \tag{10}$$

For a positive-time pole mode $s^{c-1}$, with $a + c = t - 1$ and $c \geq 1$,

$$\begin{aligned}\sum_{\substack{a+c=t-1\\ c\geq 1}} x^a s^{c-1} &= \sum_{c=1}^{t-1} x^{t-1-c} s^{c-1} \\ &= \boxed{\frac{s^{t-1} - x^{t-1}}{s - x}}.\end{aligned} \tag{11}$$

Define

$$\boxed{A_t^{(+)}(x;s) = \frac{s^{t-1} - x^{t-1}}{s - x}}. \tag{12}$$

If $x = s$, the expression is understood as the limit

$$A_t^{(+)}(s;s) = (t-1)s^{t-2}.$$

For $t = 1$, $A_1^{(+)} = 0$, as there is no room for a positive-time $H$-factor.

### Three-factor convolution

The negative part of $1 - \mathcal{F}_u(v, \cdot)$ contributes a first-passage mode $y^{b-1}$ with $b \geq 1$. The $H(0)$ contribution is

$$\boxed{B_t^{(0)}(x,y) = \sum_{\substack{a+b=t-1\\ b\geq 1}} x^a y^{b-1} = \frac{x^{t-1} - y^{t-1}}{x - y}}. \tag{13}$$

For the positive-time $H$-pole contribution, $a + b + c = t - 1$, $b \geq 1$, and $c \geq 1$:

$$\sum_{\substack{a+b+c=t-1\\ b\geq 1,\, c\geq 1}} x^a y^{b-1} s^{c-1} = [z^{t-3}] \frac{1}{(1-zx)(1-zy)(1-zs)}. \tag{14}$$

Define

$$\boxed{B_t^{(+)}(x,y;s) = [z^{t-3}] \frac{1}{(1-zx)(1-zy)(1-zs)}}. \tag{15}$$

When $x, y, s$ are distinct,

$$\boxed{B_t^{(+)}(x,y;s) = \frac{x^{t-1}}{(x-y)(x-s)} + \frac{y^{t-1}}{(y-x)(y-s)} + \frac{s^{t-1}}{(s-x)(s-y)}}. \tag{16}$$

Coincident cases are obtained by taking limits. This is where terms such as $t\alpha^t$ can appear in an uncompressed expansion. Such terms are not automatically true tail terms; they may cancel after the complete expression is recombined.

## 4 Corrected direct-route Eq. (41)

Substitute (6), (7), and (9) into (5), keeping the $H(0)$ part separate. The corrected long expression is

$$\boxed{\begin{aligned}
F_{n_0}(v,t) = {} & \sum_{\ell=1}^{L} f_\ell(n_0,v)\alpha_\ell^{t-1} \\
& + qh_0\left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0,u,v)\gamma_r^{t-1} + \sum_{r=1}^{L} g_r^{(2)}(n_0,u,v)\alpha_r^{t-1}\right] \\
& + q\sum_k c_k\left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0,u,v)A_t^{(+)}(\gamma_r;s_k) + \sum_{r=1}^{L} g_r^{(2)}(n_0,u,v)A_t^{(+)}(\alpha_r;s_k)\right] \\
& - qh_0\sum_{\ell=1}^{L} f_\ell(u,v)\left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0,u,v)B_t^{(0)}(\gamma_r,\alpha_\ell)\right. \\
& \qquad\qquad\qquad \left. + \sum_{r=1}^{L} g_r^{(2)}(n_0,u,v)B_t^{(0)}(\alpha_r,\alpha_\ell)\right] \\
& - q\sum_k c_k\sum_{\ell=1}^{L} f_\ell(u,v)\left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0,u,v)B_t^{(+)}(\gamma_r,\alpha_\ell;s_k)\right. \\
& \qquad\qquad\qquad \left. + \sum_{r=1}^{L} g_r^{(2)}(n_0,u,v)B_t^{(+)}(\alpha_r,\alpha_\ell;s_k)\right].
\end{aligned}} \tag{17}$$

This is the corrected direct-route replacement for the updated manuscript's Eq. (41). Compared with the printed expression, the baseline term must target $v$, the global sign inherits the positive sign in Eq. (40), and the $g^{(2)}$ modes are paired with $\alpha_r$, not with $\gamma_r$.

## 5 Why the direct route recompresses to the closed form

Now specialize to $N = 2L$, $v = 0$, $u = L$, and let $\rho$ be the folded distance from the starting site to the absorbing target. The ingredients in (2) have compact Chebyshev forms:

$$\widetilde{\mathcal{F}}_\rho(0,z) = \frac{T_{L-\rho}(y)}{T_L(y)}, \tag{18}$$

$$\widetilde{W}_\rho(L,z) = \frac{U_{\rho-1}(y)}{zq\,T_L(y)}, \tag{19}$$

$$1 - \widetilde{\mathcal{F}}_L(0,z) = 1 - \frac{1}{T_L(y)} = \frac{T_L(y)-1}{T_L(y)}, \tag{20}$$

$$\widetilde{H}(z) = \frac{T_L(y)}{aT_L(y) + U_{L-1}(y)}. \tag{21}$$

Substitution into (2) gives

$$\begin{aligned}
\widetilde{F}_\rho(z) &= \frac{T_{L-\rho}}{T_L} + qz\frac{U_{\rho-1}}{zqT_L}\frac{T_L-1}{T_L}\frac{T_L}{aT_L+U_{L-1}} \\
&= \frac{T_{L-\rho}(aT_L+U_{L-1}) + U_{\rho-1}(T_L-1)}{T_L(aT_L+U_{L-1})}.
\end{aligned} \tag{22}$$

The only remaining algebra is a Chebyshev identity:

$$T_L(y)U_{L-\rho-1}(y) = T_{L-\rho}(y)U_{L-1}(y) - U_{\rho-1}(y). \tag{23}$$

Using (23) in (22) gives exactly (1):

$$\widetilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}.$$

This proves that the closed form is the direct-route expression after the intermediate $\alpha, \gamma, s$ terms have been recombined.

# 6 Tail decay from the recompressed expression

Let

$$N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y), \qquad D(y) = aT_L(y) + U_{L-1}(y).$$

The poles of $\widetilde{F}_\rho(z)$ are the roots $D(y_k) = 0$. With $s_k = 1 - q + qy_k$, simple roots give

$$\boxed{F_\rho(t) = \sum_k B_{\rho k} s_k^{t-1}, \qquad B_{\rho k} = \frac{qN_\rho(y_k)}{D'(y_k)}.} \tag{24}$$

The long-time tail is therefore

$$\boxed{F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \qquad \frac{F_\rho(t+1)}{F_\rho(t)} \to s_1.} \tag{25}$$

It is not generically $t\alpha_1^t$.

# 7 Check of the figure in the updated PDF

The figure in the updated PDF is not completely useless. Interpreted narrowly, it is a graphical representation of the poles of $H$: the intersections of

$$\phi(1/s)$$

with the corrected horizontal line

$$-a = -\frac{q}{\beta(1-q)}.$$

That part is consistent with the corrected denominator $a + \phi(1/s) = 0$.

However, the figure is better treated as a pole-location visualization rather than as a standalone way to infer a finite transition at

$$s_1 = \gamma_1 = 1 - q + q\cos\frac{2\pi}{N}.$$

For the corrected equation,

$$D(\cos\theta) = a\cos(L\theta) + \frac{\sin(L\theta)}{\sin\theta} = 0, \qquad s_1 = 1 - q + q\cos\theta_1,$$

and the largest root satisfies

$$\boxed{\gamma_1 < s_1 < \alpha_1} \qquad (0 < \beta \le 1). \tag{26}$$

At $s = \gamma_1$, equivalently $\theta = \pi/L$, the $\phi$-term is zero, so the corrected denominator is $a > 0$, not zero. Thus $s_1 = \gamma_1$ is only approached in the infinite-shortcut-strength limit $a \to 0$, not reached for finite positive admissible $\beta$.

There is also a caption typo in the updated PDF: the sentence saying that $q > 1$ is required for a nonzero shortcut should be replaced. The shortcut mass is $\beta(1-q)$, so one needs $0 < q < 1$ and $\beta > 0$, not $q > 1$.

### Suggested replacement caption sentence

> The intersections with the horizontal line $-q/[\beta(1-q)]$ give the poles $s_k$ of the corrected auxiliary function $\widetilde{H}_{N,N/2}(z) = 1/(q/[\beta(1-q)] + \phi(z))$. This figure is a visualization of the $H$-poles only; it does not by itself determine the finite-time one-peak/two-peak transition. For finite $0 < \beta \leq 1$, the dominant corrected pole satisfies $\gamma_1 < s_1 < \alpha_1$, so the proposed finite crossing $s_1 = \gamma_1$ does not occur.

## 8 Validation summary

| Check | Result |
|---|---|
| Corrected plus/plus first-passage formula vs finite stochastic matrix | $2.08 \times 10^{-17}$ max difference |
| Corrected direct-route Eq. (17) vs finite stochastic matrix | $1.32 \times 10^{-16}$ max difference |
| Updated Eq. (32) denominator sign vs finite stochastic matrix | $3.29 \times 10^{-3}$ max difference |
| Updated Eq. (33) minus prefactor vs finite stochastic matrix | $6.34 \times 10^{-2}$ max difference |
| Closed form (1) vs finite stochastic matrix | $7.63 \times 10^{-17}$ max difference |

For $N = 100$, $q = 2/3$, and the fixed shortcut $u = 6 \to v = 56$, the clear visual double-peak scan finds clear double peaks up to about $\beta = 0.03$ for $n_0 = 1, 2, 3$, and no clear double peak at $\beta_c = 2q/[(1-q)N] = 0.04$. This supports treating bimodality as a finite-time shape question, not as a direct consequence of the asymptotic tail crossing.

## Deliverable recommendation

One useful package to send back is this direct-route note together with the short benchmark note. The direct-route note shows how the corrected long convolution is obtained and why it recompresses to the closed form; the benchmark note gives the numerical evidence for updating the remaining signs in the displayed first-passage formulae.

# Check of the Updated Shortcut Calculation

6 June 2026

## Summary

I checked the updated TeX file against the direct finite stochastic shortcut matrix. The leading corrected resolvent formula now has the stochastic shortcut sign, but the correction has not been propagated consistently into the first-passage formula. In particular, the equations labelled `first_pass_prob` and `first_pass_prob_2` still need the corresponding sign update to match the corrected resolvent and the finite-matrix benchmark.

## First-passage sign check

For a shortcut that moves probability mass

$$\lambda = \beta(1-q)$$

from the self-loop $u \to u$ to the absorbing target edge $u \to v$, the transient dynamics loses this mass at $u$. Summing the corrected absorbing propagator gives

$$\widetilde{F}_{n_0}(v,z) = \widetilde{\mathcal{F}}_{n_0}(v,z) + \frac{z\beta(1-q)\widetilde{W}_{n_0}(u,z)\left[1-\widetilde{\mathcal{F}}_u(v,z)\right]}{1+z\beta(1-q)\widetilde{W}_u(u,z)}.$$

Equivalently, with

$$\widetilde{H}_{N,|u-v|}(z) = \frac{\beta(1-q)}{q}\,\frac{1}{1+z\beta(1-q)\widetilde{W}_u(u,z)},$$

the prefactor in the next displayed first-passage equation should be positive:

$$\widetilde{F}_{n_0}(v,z) = \widetilde{\mathcal{F}}_{n_0}(v,z) + qz\widetilde{W}_{n_0}(u,z)\left[1-\widetilde{\mathcal{F}}_u(v,z)\right]\widetilde{H}_{N,|u-v|}(z).$$

As currently displayed, the first expression has a minus denominator and the second expression has a minus prefactor. These two forms therefore do not yet match the corrected $H$ definition.

## Finite-matrix benchmark

I used the direct stochastic matrix with $N = 100$, $q = 2/3$, $\beta = 0.04$, shortcut $u = 6 \to v = 56$ in paper indexing, and target $v = 56$. The check used $n_0 = 1, \ldots, 5$ and $z \in \{0.1, 0.3, 0.6, 0.85, 0.95\}$.

| Formula compared with the direct stochastic matrix | max absolute difference |
|---|---|
| correct plus/plus first-passage formula | $2.08 \times 10^{-17}$ |
| updated `first_pass_prob` denominator sign | $3.29 \times 10^{-3}$ |
| updated `first_pass_prob_2` minus prefactor | $6.34 \times 10^{-2}$ |

## Pole comparison

For the symmetric case $|u - v| = N/2$, the updated expression has

$$\widetilde{H}_{N,N/2}(z) = \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}.$$

Writing $s = 1/z$, for $N = 100$, $q = 2/3$,

$$\gamma_1 = 1 - q + q\cos\frac{2\pi}{N} = 0.9986844856,$$

and

$$\alpha_1 = 1 - q + q\cos\frac{\pi}{N} = 0.9996710402.$$

The largest pole $s_1$ solves

$$\frac{q}{\beta(1-q)} + \phi(1/s) = 0.$$

At $s = \gamma_1$, the $\phi(1/s)$ term is zero, so the denominator is $q/[\beta(1-q)] > 0$ for every finite positive $\beta$. Thus $s_1$ cannot equal $\gamma_1$ for $0 < \beta \le 1$. Numerically,

| $\beta$ | $s_1$ | $s_1 - \gamma_1$ | $\alpha_1 - s_1$ |
|---|---|---|---|
| 0.01 | 0.9996076314 | $9.23 \times 10^{-4}$ | $6.34 \times 10^{-5}$ |
| 0.04 | 0.9994512449 | $7.67 \times 10^{-4}$ | $2.20 \times 10^{-4}$ |
| 0.08 | 0.9993014753 | $6.17 \times 10^{-4}$ | $3.70 \times 10^{-4}$ |
| 1.00 | 0.9987831951 | $9.87 \times 10^{-5}$ | $8.88 \times 10^{-4}$ |

So the proposed switch at $s_1 = \gamma_1$ does not occur in the admissible range $0 < \beta \le 1$ for this corrected $H$.

## Double-peak scan

Using the exact first-passage PMF for the same $N = 100$, $q = 2/3$, $u = 6 \to v = 56$, target $v = 56$ setup, the result depends on what is meant by "double peak." With the report's clear visual rule

$$h_v / \min(h_1, h_2) \le 0.8,$$

with two filtered peaks and peak separation at least 10, the scan gives:

| $n_0$ | clear double peak? | largest checked $\beta$ with clear double peak |
|---|---|---|
| 1 | yes | 0.030 |
| 2 | yes | 0.030 |
| 3 | yes | 0.030 |
| 4 | no | – |
| 5 | no | – |
| 6 | no | – |

At $\beta_c = 2q/[(1-q)N] = 0.04$, no case remains clearly double-peaked under this rule; for $n_0 = 1, 2, 3$ the inter-peak valley is already shallow,

$$h_v / \min(h_1, h_2) \approx 0.887\text{–}0.891.$$

If one only asks for two local maxima, the boundary is less strict and local maxima persist to about $\beta = 0.060$–$0.062$ for $n_0 = 1, 2, 3$, but the shape is no longer a clear two-peak distribution.